

Shell Scripting - Complete Code for 13 Assignments (Updated)

1. Print Hello

```
echo "Hello"
```

2. Print Hello

```
echo "Enter your name:"
read name
echo "Hello $name"
```

3. Arithmetic Operations

```
echo "Enter two numbers"
read num1 num2
echo "Addition:"
expr $num1 + $num2
echo "Subtraction:"
expr $num1 - $num2
echo "Multiplication:"
expr $num1 \* $num2
echo "Division:"
expr $num1 / $num2
```

4. Sum and Average

```
echo "Enter three numbers"
read n1 n2 n3
sum=`expr $n1 + $n2 + $n3`
echo "Sum = $sum"
average=`expr $sum / 3`
echo "Average = $average"
```

5. Floating Point Addition using bc

```
echo "Enter two floating-point numbers"
read n1 n2
echo "$n1 + $n2" | bc
```

6. String Length

```
echo "Enter a string"
read str
l=`echo "$str" | wc -c`
len=`expr $l - 1`
echo "Length of \"$str\" is $len"
```

7. Store Numbers and Sort

```
echo "How many numbers do you want to enter?"
read n
> numbers.txt
i=1
while [ $i -le $n ]
do
    echo "Enter number $i"
    read num
    echo $num >> numbers.txt
    i=`expr $i + 1`
done
echo "Numbers in ascending order"
sort -n numbers.txt
```

8. Command Line Sum

```
sum=0
while [ $# -gt 0 ]
do
    sum=`echo "$sum+$1" | bc`
    shift
done
echo "Sum = $sum"
```

9. Menu Driven Calculator using case

```
echo "Enter first number"
read n1

echo "Enter second number"
read n2

echo "1.Addition"
echo "2.Subtraction"
echo "3.Multiplication"
echo "4.Division"

read choice

case $choice in
1)
    echo "Sum = `expr $n1 + $n2`"
    ;;
2)
    echo "Difference = `expr $n1 - $n2`"
    ;;
3)
    echo "Product = `expr $n1 \* $n2`"
    ;;
4)
    echo "Division = `expr $n1 / $n2`"
    ;;
*)
    echo "Invalid Choice"
    ;;
esac
```

10. until Loop

```
i=1
until [ $i -gt 5 ]
do
    echo $i
    i=`expr $i + 1`
done
```

11. if-elif-else

```
echo "Enter a number"
read num
if [ $num -gt 100 ]
then
    echo "$num is greater than 100"
elif [ $num -lt 100 ]
then
    echo "$num is less than 100"
else
    echo "$num is equal to 100"
fi
```

12. File Exists

```
echo "Enter file path"
read filepath
if [ -f "$filepath" ]
then
    cat "$filepath"
else
    echo "File does not exist"
fi
```

13. Password Validation

```
echo "Enter Password"
read pass
echo "$pass" | grep "^[A-Za-z][A-Za-z0-9_-]\{0,7\}$"
if [ $? -eq 0 ]
then
    echo "Valid Password"
else
    echo "Invalid Password"
fi
```